```
Dataset Loaded Successfully
Dataset Shape: (500, 9)

Dataset Info
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 500 entries, 0 to 499
Data columns (total 9 columns):
 #   Column          Non-Null Count  Dtype
---  ------          --------------  -----
 0   patient_id      500 non-null    int64
 1   age             450 non-null    float64
 2   gender          450 non-null    object
 3   region          450 non-null    object
 4   bmi             450 non-null    float64
 5   blood_pressure  500 non-null    float64
 6   cholesterol     452 non-null    float64
 7   glucose         450 non-null    float64
 8   disease_risk    500 non-null    int64
dtypes: float64(5), int64(2), object(2)
memory usage: 35.3+ KB
None

Statistical Summary
       patient_id         age         bmi  blood_pressure  cholesterol     glucose  disease_risk
count  500.000000  450.000000  450.000000      500.000000   452.000000  450.000000    500.000000
mean   250.500000   50.495556   26.023430      120.908640   205.674211  114.338602      0.528000
std    144.481833   17.309735    7.202912       15.452141    56.779690   45.457528      0.499715
min      1.000000   20.000000   10.760000       76.180000    79.220000   30.930000      0.000000
25%    125.750000   36.250000   22.350000      110.040000   176.267500   90.850000      0.000000
50%    250.500000   51.000000   25.335000      120.895000   201.355000  110.610000      1.000000
75%    375.250000   65.750000   28.625000      130.885000   228.342500  130.507500      1.000000
max    500.000000   79.000000   76.378667      159.030000   603.088667  442.818222      1.000000
```

```
Missing Value Summary
                Missing Values  Percentage
patient_id                   0         0.0
age                         50        10.0
gender                      50        10.0
region                      50        10.0
bmi                         50        10.0
blood_pressure               0         0.0
cholesterol                 48         9.6
glucose                     50        10.0
disease_risk                 0         0.0

Numerical Missing Values Filled Using Mean
Categorical Missing Values Filled Using Most Frequent
Missing Indicator Column Created
KNN Imputation Applied
MICE Imputation Applied

Dataset Shape After Z-Score Outlier Removal: (485, 10)
Dataset Shape After IQR: (485, 10)
Dataset Shape After Percentile Method: (490, 10)
Winsorization Applied to Cholesterol

Final Dataset Shape: (500, 10)

Final Dataset Summary
```

df.partition(

	patient_id	age	bmi	blood_pressure	cholesterol	glucose	disease_risk	bmi_missing
count	500.000000	500.000000	500.000000	500.000000	500.000000	500.000000	500.000000	500.0
mean	250.500000	50.495556	26.023430	120.908640	202.615624	114.338602	0.528000	0.0
std	144.481833	16.419628	6.832522	15.452141	32.795093	43.119996	0.499715	0.0
min	1.000000	20.000000	10.760000	76.180000	139.810000	30.930000	0.000000	0.0
25%	125.750000	38.750000	22.602500	110.040000	180.592500	93.445000	0.000000	0.0
50%	250.500000	50.495556	25.990000	120.895000	205.674211	114.338602	1.000000	0.0
75%	375.250000	63.000000	28.252500	130.885000	224.687500	128.285000	1.000000	0.0
max	500.000000	79.000000	76.378667	159.030000	264.490000	442.818222	1.000000	0.0

Final Clean Dataset Saved

Project Completed Successfully